Supplementary Appendix A. Trial-time Implementation of the GPT-4o / LungDiag-derived Mini-program Intervention

This appendix provides a full operational specification of the intervention that was deployed and tested during the trial. It is written so that an independent team could reproduce the system at the architectural and workflow level, while commercially sensitive assets (exact prompt text, proprietary retrieval-engine configuration, and the LungDiag-derived knowledge layer) remain undisclosed. Where disclosure is withheld, we give a functionally equivalent mock specification sufficient for appraisal.

A1. Architecture overview

The evaluated intervention is an integrated system with four layers (Figure A1):

1. **Client layer** — a WeChat mini-program that renders the structured questionnaire, accepts Chinese text input, and displays task-constrained output.
2. **Orchestration layer** — a stateless task router (Xingyun-hosted) that enforces the three-task workflow, selects the matching prompt template, enforces input/output guardrails, and writes audit logs.
3. **Retrieval layer** — a retrieval-augmented generation (RAG) stack that queries a frozen LungDiag-derived respiratory knowledge layer and returns the top-k respiratory-domain chunks.
4. **Generation layer** — GPT-4o (OpenAI), accessed through APIYI-routed API under commercial paid terms, invoked with task-specific prompts plus retrieved evidence.

The reference open-source stack used to rebuild an equivalent pipeline is:

- **LangChain** (https://github.com/langchain-ai/langchain) or **LlamaIndex** (https://github.com/run-llama/llama_index) for orchestration and retriever composition.
- **BGE-M3** (https://github.com/FlagOpen/FlagEmbedding) for Chinese-capable dense + sparse embeddings.
- **FAISS** (https://github.com/facebookresearch/faiss) or **Milvus** (https://github.com/milvus-io/milvus) for the vector store.
- **bge-reranker-v2-m3** for cross-encoder reranking.
- Alternative end-to-end reference: **LightRAG** (https://github.com/HKUDS/LightRAG) for graph-augmented retrieval; this matches the "knowledge-graph reasoning" terminology used in the main text and was used during internal prototyping to validate the retrieval topology.

The deployed trial version is a proprietary adaptation of the above topology. Any re-implementation built from these open-source components against a comparable respiratory knowledge layer will reproduce the functional behavior evaluated in the trial.

A2. LungDiag-derived respiratory knowledge layer

A2.1 Provenance

The knowledge layer is derived from the LungDiag respiratory diagnostic framework (Liang et al., *MedComm* 2025;6:e70043), which was itself built from 31,267 de-identified electronic health records (EHRs) from the National Center for Respiratory Medicine and affiliated centers (admission dates 1 Nov 2012 – 30 Oct 2019) and externally tested on 1,142 EHRs from three additional hospitals. The original framework uses a Bi-LSTM-CRF named-entity recognizer with BIO tagging, the PIAT interactive annotation platform, UMLS/ICD-10 concept standardization, and a fine-grained deep-phenotyping pipeline that yielded 252 standardized clinical features across 10 respiratory disease categories (COPD, bronchial asthma, bronchiectasis, airway stenosis, pulmonary hypertension, lung space-occupying lesions / lung cancer, pulmonary infectious diseases, pulmonary tuberculosis, pleural disease, interstitial lung disease).

A2.2 Scope, form, and transformations applied for the trial

For the present trial, the knowledge layer was re-engineered from the LungDiag feature graph into a retrievable document corpus. No raw identifiable EHR text was indexed. The indexed content consists of:

- **Disease concept cards** (n ≈ 10 major categories + subtypes): canonical name, ICD-10 codes, typical chief complaints, cardinal and supporting symptoms, imaging patterns, laboratory signatures, first-line medications, common differentials.
- **Symptom → disease association records** (n ≈ 250 standardized features × weighted edges to the 10 disease categories), serialized as short evidence statements ("productive cough > 3 months in ≥ 2 consecutive years is associated with COPD…").
- **Triage-pattern records**: red-flag symptom clusters mapped to the four trial triage levels (emergency, within one day, within one week, self-care), derived from category-level EHR outcomes and cross-referenced against the respiratory terminology base of 11,988 clinical terms released with the original LungDiag study.

Questionnaire cases, gold-standard answers, and scoring rubrics were held out of the index and stored in a separate audit-only repository (Appendix B, Appendix C); no chunk in the retrieval index contains trial case text, trial diagnostic keys, or triage keys.

A2.3 Chunking and indexing

Each record was segmented into semantic chunks of 256–512 tokens with a 64-token overlap using a recursive-character splitter with Chinese sentence-boundary heuristics (equivalent to LangChain's RecursiveCharacterTextSplitter configured for Chinese punctuation). Chunks carry the following metadata: doc_id, disease_category, content_type ∈ {disease_card, symptom_association, triage_pattern}, source_version. Embeddings were produced with a BGE-family Chinese embedding model; a parallel sparse index (BM25) was built for hybrid retrieval. The vector store was hosted inside the Tier-III-classified environment; chunks never leave the deployment perimeter.

A2.4 Guideline relationship

Guideline materials from the Chinese Medical Association Respiratory Disease Branch, GOLD, and ATS/ERS were used only for questionnaire construction and Delphi gold-standard adjudication (see Methods, Clinical Case Construction). These guideline documents were **not** indexed into the trial-time knowledge layer. This separation was verified by (i) hashing every document in the knowledge-layer corpus, (ii) hashing every questionnaire case and gold-standard answer, and (iii) confirming no hash overlap before freeze.

A2.5 Governance, freeze, and update policy

The knowledge layer is owned and maintained by the authors' research group. Vendors (Xingyun, APIYI) had no role in content governance, curation, or updates. The trial-time release (internal version V1.0) was frozen on 31 March 2025, before participant enrollment, and remained fixed throughout enrollment (April–June 2025). No content edits, re-embeddings, or schema changes were performed during enrollment. Post-trial updates follow a separate governance cycle outside the scope of this report.

A2.6 Retrieval-quality management

No standalone retrieval benchmark was prespecified as a primary outcome. Retrieval relevance was managed as a trial-time quality-control process:

- **Pre-deployment checks**: 200 synthetic respiratory queries (stratified over the 10 disease categories and three task types) were executed; the primary acceptance criterion was that ≥ 1 of the top-5 retrieved chunks carried metadata matching the ground-truth disease category (in internal checks the observed rate was ≥ 0.9; the exact figure is recorded in the internal acceptance report).
- **Audit logging during trial**: every retrieval call wrote retrieval_status, retrieved_doc_ids, and retrieval_score_top1 to the audit log (Section A7), enabling post hoc inspection of retrieval behavior without altering the deployed logic.

A3. Trial-time workflow

For every participant–item pair the workflow was:

[Participant text in mini-program]

|
▼

[Input guardrail] —— off-scope → refusal template (Section A6)

|
▼

[Task router] — selects one of {T1 risk/etiology, T2 diagnosis, T3 triage}

|
▼

[Query builder] — task-specific query reformulation

|
▼

[Hybrid retrieval: BM25 + dense (BGE-M3)]

|
▼

[Cross-encoder rerank → top-k = 5]

|
▼

[Prompt assembly: task template + retrieved evidence + user text]

|
▼

[GPT-4o call (temperature 0.2, max_tokens 512, JSON schema)]

|
▼

[Output guardrail: schema + scope check] —— fail → one retry, then safe default

|
▼

[Structured output rendered in mini-program]

|
▼
[Audit log write]
External web search, open-ended consultation, follow-up multi-turn dialogue outside the structured task, and any non-respiratory content were disabled in the deployed trial version.

A4. Prompt frameworks (mock equivalents)

The exact prompt strings are proprietary. The following mock templates are functionally equivalent at the task and structure level and can be used by an independent team to reproduce the workflow. Placeholders use {{variable}} syntax.

A4.1 System prompt (shared)

You are a clinical-information assistant constrained to the respiratory disease domain. You answer only three task types: (T1) risk-behavior or etiologic clue identification, (T2) preliminary diagnosis, (T3) triage recommendation.

Rules:
- Use only the information in {{retrieved_evidence}} and the user's case text.
- If the user's question is not respiratory, reply with the refusal template.
- Return a single JSON object matching the schema for the current task.
- Do not provide treatment plans, drug dosing, or procedural instructions.
- Do not invent evidence not present in {{retrieved_evidence}}.
- All responses are in simplified Chinese.

A4.2 Template 1 — Risk-behavior / etiology identification (T1)

Task: Identify the single highest-risk behavior or etiologic clue implied by the case.

Case text:
{{user_case_text}}

Retrieved respiratory-domain evidence (top-k, frozen knowledge layer):
{{retrieved_evidence}}

Respond as JSON:
```
{
  "task": "T1_risk",
  "risk_factor": "<short noun phrase, in Chinese>",
  "evidence_chunk_ids": ["<id1>", "<id2>", ...],
  "confidence": "<high|medium|low>"
}
```

A4.3 Template 2 — Preliminary diagnosis (T2)

Task: Select the single most likely preliminary respiratory diagnosis from the case. Restrict the answer to respiratory conditions supported by the retrieved evidence.

Case text:
{{user_case_text}}

Retrieved respiratory-domain evidence (top-k, frozen knowledge layer):
{{retrieved_evidence}}

Respond as JSON:
{
"task": "T2_diagnosis",
"preliminary_diagnosis": "<single Chinese disease term>",
"supporting_features": ["<feature 1>", "<feature 2>", ...],
"evidence_chunk_ids": ["<id1>", "<id2>", ...],
"confidence": "<high|medium|low>"
}

A4.4 Template 3 — Triage recommendation (T3)

Task: Produce a triage recommendation from the four trial categories: "emergency", "within one day", "within one week", "self-care". Base the recommendation on symptom severity, red-flag features, and the retrieved evidence.

Case text:
{{user_case_text}}

Retrieved respiratory-domain evidence (top-k, frozen knowledge layer):
{{retrieved_evidence}}

Respond as JSON:
{
"task": "T3_triage",
"triage_level": "<emergency|within_one_day|within_one_week|self_care>",
"red_flags": ["<flag 1>", "<flag 2>", ...],
"rationale": "<one-sentence Chinese justification>",
"evidence_chunk_ids": ["<id1>", "<id2>", ...]
}

A4.5 Refusal template (out-of-scope)

{
"task": "refusal",
"reason": "该问题不在本工具的呼吸系统评估范围内；请咨询相关专科医生。"
}

The production system uses internally optimized Chinese prompt strings with additional safety language. The mock templates above preserve the **task boundary, required output schema, evidence-grounding requirement, and refusal behavior** that an independent evaluator needs to reproduce the system's behavioral envelope.

A5. Retrieval and generation configuration (functional specification)

Parameter	Deployed trial value	Reference open-source equivalent
Chunk size	256–512 tokens, 64-token overlap	RecursiveCharacterTextSplitter (LangChain)
Embedding model	BGE-family Chinese dense model	BAAI/bge-m3 (HuggingFace)
Sparse index	BM25 over segmented Chinese text	rank_bm25 + jieba tokenizer
Hybrid fusion	Reciprocal-rank fusion	LlamaIndex QueryFusionRetriever
Reranker	Cross-encoder, Chinese-capable	BAAI/bge-reranker-v2-m3
Top-k retrieved	5	—
Domain filter	metadata = respiratory	Vector-store metadata filter
LLM	GPT-4o (OpenAI)	—
Temperature	0.2	—
max_tokens	512	—
Output format	JSON schema enforced	response_format={"type":"json_object"}
Retry policy	1 retry on schema failure, then safe default	—

Exact chunking parameters, reranker thresholds, domain filters, and query-rewrite rules used in the deployed version are proprietary. The values above are documented at the resolution needed to reproduce functional behavior; fine-tuning within the specified ranges does not change the trial-time task boundaries.

A6. Guardrails

Three guardrail layers were active throughout enrollment:

1. **Input guardrail** — a keyword + classifier check rejected questions outside the respiratory domain, queries requesting drug dosing or surgical instructions, and queries requesting identification of individuals. Rejections returned the refusal template (A4.5) and were logged with error_flag = "out_of_scope".
2. **Retrieval guardrail** — queries returning zero respiratory-domain chunks above a minimum similarity threshold triggered a "low-evidence" path: the system returned the refusal template rather than generating unsupported output, and logged retrieval_status = "low_evidence".
3. **Output guardrail** — the generated JSON was validated against the task's schema; unknown fields, missing required fields, or out-of-vocabulary values (e.g., triage level not

in the four allowed strings) caused one automatic retry with a stricter reminder, then — if the retry also failed — a safe default response was returned (refusal for T1/T2; "within_one_week" with an explicit uncertainty note for T3) and error_flag = "schema_fail" was recorded.

The deployed system does **not** fall back to unrestricted GPT-4o responses, external web search, or cached answers when a guardrail trips.

A7. Logging fields

Every participant–item call wrote a single audit record with the following fields:

Field	**Description**	**Use**
timestamp_utc	ISO-8601 server timestamp	Audit trail
session_id	Salted hash of participant/session identifier	De-identified linkage
assigned_task_type	T1 / T2 / T3	Workflow trace
user_input_text	Verbatim Chinese input	Audit + post hoc review
retrieval_status	ok / low_evidence / error	QC
retrieved_doc_ids	List of top-k chunk IDs (hashed references)	Retrieval audit
retrieval_score_top1	Top-1 rerank score	Retrieval audit
ai_output_text	Raw JSON string returned by GPT-4o	Audit + scoring support
error_flag	none / out_of_scope / low_evidence / schema_fail / api_error	Failure mode tracing
completion_time_ms	Wall-clock turn latency	Secondary outcome
model_version	GPT-4o model tag reported by API	Reproducibility
knowledge_layer_version	V1.0 (frozen 2025-03-31)	Reproducibility

Logs were read-only for investigators, reviewed for operational health only, and did not feed back into the deployed logic during enrollment.

A8. Failure modes and handling

Failure mode	Detection	Handling	Logged flag
Off-scope user input	Input guardrail	Refusal template	out_of_scope
No relevant retrieval	Score < threshold	Refusal template	low_evidence
LLM API timeout / error	API layer	One retry; then safe default response	api_error
Malformed JSON output	Output schema validator	One retry with stricter reminder; then safe default	schema_fail
Triage level outside four allowed values	Output schema validator	Same as above; safe default = within_one_week + uncertainty note	schema_fail
Mini-program / network error on client	Client telemetry	Participant re-prompted; item recorded as unanswered	—

All failure paths are **rule-bound and non-adaptive**: the deployed intervention logic was not modified mid-enrollment on the basis of any individual output.

A9. Deployment boundaries

The evaluated workflow was restricted to:

- text-only Chinese input inside the mini-program;
- retrieval from the frozen LungDiag-derived respiratory knowledge layer only;
- one of three task-constrained outputs per call;
- no external web search, no unrestricted open-ended consultation, no multi-turn follow-up outside the structured questionnaire, no image or voice input, no connection to patient EHRs.

Features available in the broader product environment but not enabled during the trial are out of scope for this evaluation.

A10. Availability, reproducibility, and non-disclosure boundary

Released (DOI-backed, citable):

- Statistical analysis code: https://github.com/asleepm/rct-analysis-code/releases/tag/v1.2
- Supplementary Appendix A (this document), Appendix B (scoring rubric and questionnaire administration), Appendix C (composition of the 88-case respiratory question bank), Supplementary Data 1 (triage urgency source data).

Released to reviewers only: full questionnaire materials and gold-standard answer keys.
Not released (commercial / data-governance constraints):

- Trial-specific WeChat mini-program source code.
- Exact internal prompt text, exact chunking and reranker thresholds, proprietary retrieval-engine configuration.
- The LungDiag-derived respiratory knowledge layer itself.

Minimal reproducible specification provided instead (this appendix): architecture, knowledge-layer provenance/scope/governance/freeze policy, chunking and embedding approach with open-source equivalents, mock prompt templates preserving the task/schema envelope, guardrail layers, logging fields, failure-mode table, and deployment boundaries. An independent team building a system against the above specification with the open-source stack listed in Section A1 can reproduce the functional behavior evaluated in this trial.

Supplementary Table A1. LungDiag-derived respiratory knowledge layer (trial-time summary)

Attribute	Value
Trial-time role	Evidence-support layer for RAG; not a case bank, answer-key repository, or scoring rubric
Provenance	Derived from LungDiag EHR-based respiratory framework (Liang et al., 2025)
Data form	Standardized concept cards, symptom–disease associations, triage patterns; no raw identifiable EHR text
Clinical scope	Respiratory disease-related content only
Terminology scope	Disease names, symptoms, laboratory tests, imaging findings, medications, procedures
Disease categories	10 original LungDiag categories
Guideline relationship	Guideline and questionnaire materials maintained separately; not indexed
Chunk size	256–512 tokens, 64-token overlap
Embedding	BGE-family Chinese model
Retrieval	Hybrid dense + BM25, cross-encoder rerank, top-k = 5
Trial version	Frozen internal release V1.0
Freeze date	31 March 2025 (before enrollment)
Update policy	Fixed throughout enrollment (April–June 2025)
Governance	Authors' research group; Xingyun/APIYI had no governance role

Attribute	Value
Leakage prevention	Hash-based separation of index from questionnaire/answer keys
Retrieval QC	Pre-deployment acceptance checks + trial-time audit logging
Interpretation boundary	Evaluates the integrated mini-program, not the knowledge layer in isolation

Supplementary Table A2. Trial-time logging categories

See Section A7 above. All fields were captured for every participant–item call.